1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}}\,\mathrm{d}x$$

giving your answer to 2 decimal places.

(3)

(b) Using your answer to part (a) and making your method clear, estimate

(i) $\displaystyle\int_5^7 2^{6-\sqrt{x}}\,\mathrm{d}x$

(ii) $\displaystyle\int_5^7 \left(3 + 2^{5-\sqrt{x}}\right)\mathrm{d}x$

(4)